

BIOE242 Final Project Goal Write-Up – Christian Chung

My final project takes inspiration from my current PhD thesis work. Broadly, my work investigates model development within computational tissue engineering, with a greater focus on the mechanical and environmental cues that dictate tissue development. Specifically, I am looking at epithelial tissue development as they exhibit rheological behaviors (solid/fluid-like behaviors and phase transitions) that are dependent on mechanical properties. To accomplish this, I am using vertex models as the standard computational framework, where cells are modeled as polygons and energy is computed using the Farhadifar model.

In my field, two key challenges exist with tissue development: 1) being able to translate simulations to the wet lab in an accurate manner, and 2) the bottlenecks with time and cost. However, with advances in machine learning, I believe that we can accomplish both of these goals to both make our simulations grounded in realism and further make the approach efficient.

My proposal is that by using methods in machine learning from this course, we can determine if cell topology alone can encode the tissue's mechanical state (specifically, its rheological behavior). Data from this lab is created by my current existing simulation pipelines, where I am able to produce vtu files/mesh snapshots from the simulations (simulations can run for many steps, and I often record a “snapshot” every 100 steps for a single simulation). Furthermore, we can vary the data by implementing parameters such as cortical tension alterations, varied proliferation rates, and altered cell-cell adhesion.

My goals for this project have three key components: 1) clustering tissues by feature vectors (to recover known solid/fluid phase boundaries), 2) predicting per-cell stress or global tissue energy from graph topology, and 3) comparing topology-aware vs topology-blind systems to quantify information gain from message passing. From an unsupervised model perspective, I intend to use **PCA + UMAP** (to reveal dominant axes of variations as cell features are inherently correlated) and **K-Means** (setting $K=2$ to classify solid and fluid tissues but also extend to $K=3-5$ to see if sub-phases exist). From a supervised model perspective, I plan on using **graph neural networks** as I am able to extract out nodes and features very easily from vertex model snapshots (cells as nodes, vertices between them as edges) to understand how message passing propagates neighborhood information.

In terms of data availability, each graph will have anywhere from 50-200 nodes minimum, with 25k-200k individual node predictions from a singular snapshot (and given the scale of how I can produce these simulations, data will not be a concern in this project). To establish a null hypothesis, I also intend on using a MLP on graph-aggregated scalars as the null to demonstrate that GNNs can outperform the MLP (helping to support the claim that topology adds information). In terms of any concerns with snapshots not being independent (ie. tissues looking the same at close time points), we will split the procedure by simulation and not by snapshot to prevent temporal autocorrelation (example hypothetical: if I run 10 proliferation, 10 adhesion, and 10 tension simulations with altered variations, I will do a training/validation/testing split ratio of 7/2/1 so that we train on 21 simulations, validate on 6, and test on 3; I will do more than this in my actual project).